MINIMUM VIABLE HUMANOID · COMPONENT LIST AND COST ESTIMATE

JX0

walks · talks · 17 STS3215 joints



46.0 cm	2.88 kg	17	65	29
HEIGHT	MASS	STS3215 JOINTS	PRINTED PARTS	BOM LINES

PARTS FOR ONE ROBOT

₹57,878

THE 17 JOINT SERVOS

₹41,123

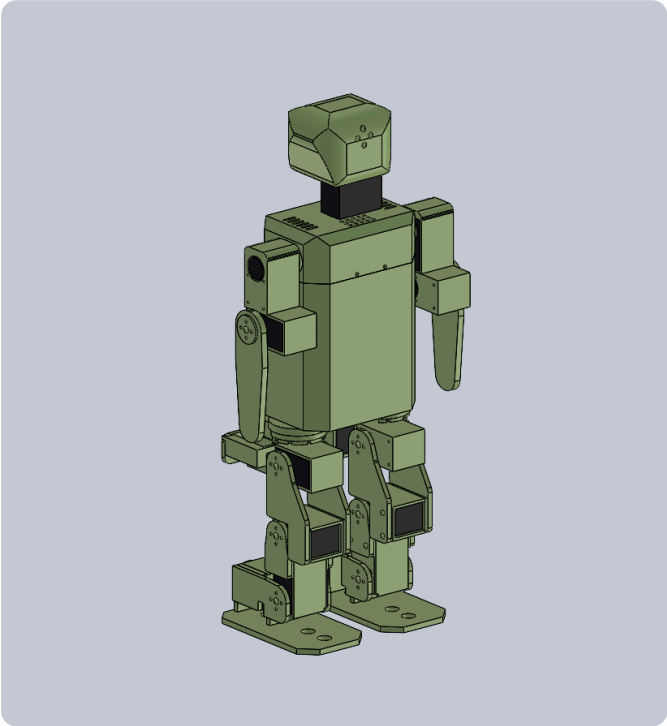
24 Sep – 5 Oct 2026 · all figures from the JX0 bill of materials (github.com/nickthelegend/thenar-jx1/tree/main/jx0)

1. What JX0 is

JX0 is a small humanoid robot designed to be built by one student at home, from a 3D printer and parts that can be ordered online in India. Its look and its joints follow a friend's working robot: a faceted green body, every joint a 12 V STS3215. It is the working prototype of the JX1 project: it uses the same design pipeline (gait planner, servo sizing, physics simulation) at a price a student can afford.

Specification

Height / mass	46.0 cm / 2.88 kg (CAD, all parts)
Joints	17, every one a Feetech STS3215 12 V serial bus servo (30 kg·cm stall, position feedback), like the reference robot
Legs	6 joints each; every pitch and roll joint double-sided (a U-bracket on the servo's drive horn and passive horn, the body in a cage), like the reference robot; the hip-yaw disc held from both sides; every part fitted to the real STS3215
Arms and head	shoulder pitch and elbow per arm, a paddle blade; neck yaw under a soft rounded head
Brain	Raspberry Pi 4: 100 Hz gait playback with arm swing, IMU balance, offline speech recognition and voice
Talking	Claude over Wi-Fi writes the replies and calls the robot's actions (walk, turn, wave, nod, look)
Power	3S 2200 mAh LiPo (about 73 min of walking, CALCULATED) or a 12.6 V bench supply
Structure	65 parts, 847 g on a home 3D printer: pelvis and leg parts in PETG, printed near-solid; body, head and arms in pastel green matte PLA



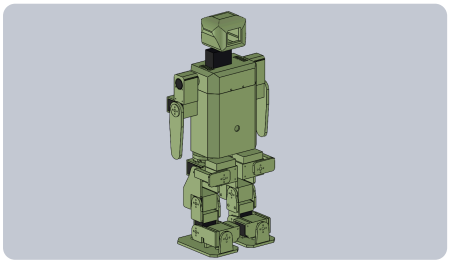
Done (on the computer)

- ✓ SolidWorks CAD: 65 printable parts and the full assembly with all 17 servos, built by script
- ✓ Servo sizing all pass (tightest 1.46×); every printed leg part passes a finite-element check
- ✓ Walking in a physics simulation of the CAD robot: 14 of 14 gaits pass, arms swinging
- ✓ No part collisions in 1,373 moving poses; all 140 walks with random model errors and every push up to 0.96 N·s stay up
- ✓ Robot program (voice, Claude, walking, gestures) runs on the simulated robot: 24/24 missions
- → **Next: buy the parts → print → assemble → first steps**

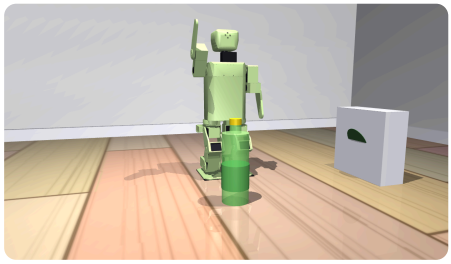
How this estimate was made

- Every line is a real product page at an Indian store (Robu, Evelta, Robocraze, ThinkRobotics, Quartz Components and others), price checked 24 Sep – 5 Oct 2026.
- GST is included where the store states it; shipping is not included.
- Four small lines (filament for the pelvis and leg brackets, hip-yaw thrust rings and keepers, push-to-talk button, jumper wires and servo cables) are ESTIMATED.
- Over the original ₹50,000 target because all 17 joints are STS3215, as on the reference robot.

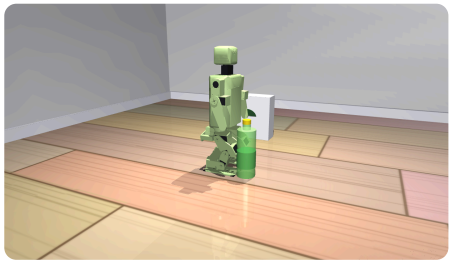
From CAD to a working robot



SolidWorks assembly, back view



waving (simulation, the robot's own program)



walks into a bottle, as in the reference video

2. Cost summary

PARTS TOTAL
₹57,878
29 line items, one robot

JOINT SERVOS
71 %
of the cost: 17 × STS3215 at ₹2,419

EVERYTHING ELSE
₹16,755
brain, power, voice, structure

By group



Groups

GROUP	WHAT	₹	SHARE
Servos	17 × STS3215 (legs, arms, neck)	₹41,123	71.1 %
Electronics	Raspberry Pi 4, microSD, servo driver, wiring	₹9,251	16.0 %
Structure	filament, screws, inserts, foot rubber	₹3,686	6.4 %
Power	LiPo, charger, 5 V converter, switch, alarm	₹3,122	5.4 %
Voice	microphone, amplifier, speaker	₹569	1.0 %
Sensors	IMU	₹127	0.2 %
Total		**₹57,878**	100 %

Five biggest lines

ITEM	MODEL	QTY	₹
Joint servo (all 17 joints: 12 legs, 4 arms, neck)	Feetech STS3215 12 V 30 kg·cm serial bus servo (sold as ST3…	17	**₹41,123**
Brain	Raspberry Pi 4 Model B 2 GB (with case)	1	**₹7,000**
Battery	Bonka 3S 11.1 V 2200 mAh 35C LiPo	1	**₹2,039**
microSD card	SanDisk Ultra 32 GB A1	1	**₹1,499**
Filament for the body, head and arms (the reference robot's pastel green)	eSun PLA-Matte 1.75 mm 1 kg, Mint Green	1	**₹1,299**

Why the STS3215 everywhere: the walking needs up to 2.0 N·m at the hip roll (with a 1.25–1.5× safety margin), more than MG996R-class hobby servos give; the serial bus servos also report their position, which the calibration and the balance loop use; and one servo type on one bus is how the reference robot is built.

3. The 17 joint servos — 71 % of the cost

Every joint is the same Feetech STS3215 12 V serial bus servo, as on the reference robot: a metal-gear servo with a 360° magnetic encoder that reports its position, load, voltage and temperature over one shared serial bus. One servo type means one spare fits every joint.

SERVOS	EACH	ALL 17
17	**₹2,419**	**₹41,123**
12 legs, 4 arms, 1 neck	Evelta, GST included	71 % of the robot

What each joint needs (simulated, with safety margins)

JOINT	SERVOS	NEEDS	OF 2.94 N·M STALL	MARGIN	SPEED USE
Hip yaw	2	0.14 N·m	**5 %**	×20.73	5 %
Hip roll	2	2.01 N·m	**68 %**	×1.46	83 %
Hip pitch	2	0.37 N·m	**13 %**	×7.99	19 %
Knee	2	1.32 N·m	**45 %**	×2.22	69 %
Ankle pitch	2	1.61 N·m	**55 %**	×1.82	25 %
Ankle roll	2	1.06 N·m	**36 %**	×2.78	45 %
Shoulder pitch	2	2.09 N·m	**71 %**	×1.4	—
Elbow	2	0.01 N·m	**0 %**	—	—
Neck yaw	1	0.00 N·m	**0 %**	—	—

Legs: the servo sizing (jx0/analysis/sizing.py) over 5 gaits and 8 static cases, dynamic peaks ×1.5 and static ×1.25; "speed use" is the share of the torque-speed line used at the fastest gait. Arms and neck: the walking peak ×1.5 (the arm swing). Every joint passes; the hip roll, holding the body over one foot, is the tightest.

Servos considered

SERVO	TORQUE	FEEDBACK	EACH	FOR 17	ENOUGH FOR THE HIP ROLL?
TowerPro MG996R	11 kg·cm = 1.08 N·m maker's figure at 6 V	none (PWM)	₹259–409	₹4,403–6,953	no (54 %)
Waveshare ST3215 7.4 V	19.5 kg·cm = 1.91 N·m at 7.4 V	position, load, temperature	₹2,159	₹36,703	no (95 %)
Hiwonder LX-224	20 kg·cm = 1.96 N·m listing	position	₹2,900	₹49,300	no (97 %)
Waveshare ST3020	25 kg·cm = 2.45 N·m listing	position, load, temperature	₹2,618	₹44,506	yes, ×1.22
Feetech STS3215 12 V (chosen)	30 kg·cm = 2.94 N·m at 12 V	position, load, temperature	₹2,419	₹41,123	yes, ×1.46
Waveshare ST3025	40 kg·cm = 3.92 N·m listing	position, load, temperature	₹10,299	₹1,75,083	yes, ×1.95

The hip roll needs 2.01 N·m with its margin. MG996R-class hobby servos are a tenth of the price but have half the torque and no feedback (the balance loop and the calibration need to read the joint angles). The 7.4 V ST3215 and the LX-224 fall just short. The ST3020 would work but costs more per joint; the ST3025 is over four times the price.

4. Printed structure and hardware

25 printed parts and 40 thin horn shims. The pelvis and every leg part carry the robot's weight at every step, so they are PETG printed near-solid (6 walls, 6 top and bottom layers, 40 % gyroid); the body, head and arms are matte PLA in the reference robot's pastel green.

PART	QTY	MATERIAL	EACH	ALL	FILAMENT ₹	FEA SF FATIGUE / STATIC
Pelvis	1	PETG near-solid	73.0 g	73 g	₹55	2.3 / 4.6
Hip-yaw keeper	2	PETG near-solid	2.9 g	6 g	₹4	2.2 / 4.8
Hip-yaw ring insert	2	PETG near-solid	0.7 g	1 g	₹1	—
Hip-yaw bracket (disc)	2	PETG near-solid	50.9 g	102 g	₹77	2.1 / 2.9
Hip-roll U-bracket	2	PETG near-solid	30.5 g	61 g	₹46	3.6 / 5.0
Thigh	2	PETG near-solid	42.0 g	84 g	₹64	2.8 / 3.8
Shin	2	PETG near-solid	23.7 g	47 g	₹36	2.7 / 3.1
Ankle bracket	2	PETG near-solid	37.7 g	75 g	₹57	2.2 / 3.0
Foot	2	PETG near-solid	50.8 g	102 g	₹77	4.7 / 6.4
Torso (lower shell, skirt)	1	PLA matte	103.0 g	103 g	₹134	—
Chest cap	1	PLA matte	45.3 g	45 g	₹59	—
Head	1	PLA matte	60.2 g	60 g	₹78	—
JX0_NeckHub	1	PLA matte	10.7 g	11 g	₹14	—
Upper arm (hood + elbow box)	2	PLA matte	21.5 g	43 g	₹56	—
Arm blade	2	PLA matte	14.8 g	30 g	₹38	—
Horn shim, 0.4 mm	40	PLA matte	0.1 g	4 g	₹5	—
Printed parts		PETG 551 g, PLA 296 g		**847 g**	**₹803**	

Filament ₹ = the part's share of the 1 kg spools (PETG ₹759, PLA ₹1,299). FEA SF: finite-element safety factors under the simulated walking (fatigue) and the hardest pushes (strength); 2 or more everywhere.

Filament, screws and small parts

#	ITEM	STORE	QTY	UNIT	TOTAL
21	**Filament for the body, head and arms (the reference robot's pastel green)** eSun PLA-Matte 1.75 mm 1 kg, Mint Green	Robocraze	1	₹1,299	**₹1,299**
21b	**Filament for the pelvis and leg brackets** PETG 1.75 mm 1 kg, green (Quartz Components PETG Pro)	Quartz Components · ESTIMATED	1	₹759	**₹759**
22–23	**Screws** M2 CSK 6 / 8 / 10 / 12 mm, M3 × 8 socket and pan head (verify_fit)	local fastener supplier	245	—	**₹520**
24	**M3 heat-set inserts** M3 x 4 brass heat-set inserts	local fastener supplier	25	₹4	**₹100**
25	**Foot grip** Natural rubber sheet 1 mm x 100 mm x 1 m	IndustryBuying	1	₹978	**₹978**
25b	**Hip-yaw thrust rings and keepers** PTFE thread-seal tape, 1 roll	any hardware shop · ESTIMATED	1	₹30	**₹30**

Structure **₹3,686**

5. Electronics, sensors and voice

A Raspberry Pi 4 runs everything on the robot: it plays the verified gaits 100 times a second with the IMU's balance corrections, listens with an offline speech recogniser, asks Claude over Wi-Fi what to say and do, and speaks through a small amplifier and speaker.

#	ITEM	STORE	QTY	UNIT	TOTAL
03	**Bus servo driver** Waveshare Serial Bus Servo Driver board	Robu	1	₹432	**₹432**
04	**Brain** Raspberry Pi 4 Model B 2 GB (with case)	ThinkRobotics	1	₹7,000	**₹7,000**
05	**microSD card** SanDisk Ultra 32 GB A1	Quartz Components	1	₹1,499	**₹1,499**
06	**IMU** MPU6050 gyro + accelerometer	Quartz Components	1	₹127	**₹127**
07	**Microphone** INMP441 I2S MEMS microphone	Quartz Components	1	₹145	**₹145**
08	**Amplifier** MAX98357A I2S 3 W class-D amplifier	Quartz Components	1	₹124	**₹124**
09	**Speaker** Waveshare 2030 cavity speaker 8 Ω 2 W	ThinkRobotics	1	₹300	**₹300**
26	**Push-to-talk button** 12 mm momentary push button	any electronics shop · ESTIMATED	1	₹20	**₹20**
27	**Jumper wires and servo cables** Dupont jumpers + 4 STS/SCS servo cables, 5264 3-pin, 30 cm (Fee…	any electronics shop · ESTIMATED	1	₹300	**₹300**

Electronics, sensors and voice **₹9,947**

How it connects

Raspberry Pi 4 — gaits + balance · 100 Hz
USB →
Servo driver — serial bus, 1 Mbit/s
1 bus →
17 × STS3215 — IDs 1–17 · angle, load, temperature

MPU6050 IMU → Pi · I2C

INMP441 microphone → Pi · I2S

Pi · I2S → MAX98357A → speaker

push-to-talk button → GPIO

Control-loop budget

Loop period	10 ms (100 Hz)
Servo bus per frame	1.27 ms (127 bytes)
IMU read	0.4 ms (estimated)
Robot program on a Pi 4	0.12 ms (estimated)
Used of each frame	**1.79 ms of 10 ms**

Talking

- Speech recognition (Vosk) and the voice (Piper) run on the Pi, offline and free.
- Claude writes the replies and chooses the robot's actions; it needs Wi-Fi and an Anthropic API key, paid per use (not in this estimate).
- The microphone listens through the four holes in the head, the speaker sits under the chest grille.

6. Power and battery life

The 12 V servos run straight from a 3S LiPo (11.1 V nominal, 12.6 V full), like the reference robot on its 12.6 V bench supply; a 5 V UBEC feeds the Pi. The same XT60 plug takes a bench supply for long sessions at the desk.

#	ITEM	STORE	QTY	UNIT	TOTAL
12	**Battery** Bonka 3S 11.1 V 2200 mAh 35C LiPo	Robocraze	1	₹2,039	**₹2,039**
13	**Charger** B3 balance charger for 2S/3S LiPo	Quartz Components	1	₹294	**₹294**
14	**5 V supply for the Pi** UBEC 5 V 5 A (2-8S input)	Robu	1	₹409	**₹409**
16	**Battery connector** XT60 male/female pair	Robocraze	2	₹29	**₹58**
17	**Power switch** 10 A DC-rated toggle switch (CALONIX)	Quartz Components	1	₹63	**₹63**
18	**Power wire** 16 AWG silicone wire (per metre)	Robu	2	₹49	**₹98**
19	**Signal wire** 20 AWG silicone wire (per metre)	Robu	3	₹21	**₹63**
20	**Battery alarm** 1-8S LiPo voltage checker / low-voltage buzzer	Quartz Components	1	₹98	**₹98**

Power	₹3,122

Battery life (calculated)

Pack	3S 2200 mAh 35C · 24.4 Wh
Walking, average	1.45 A (16 W, servos + Pi)
Walking, peak	2.52 A (pack limit 77 A)
Walking time	**about 73 minutes**
Standing time	about 127 minutes
Reference robot	0.6–1.5 A at 12.6 V walking (its video)

From the simulated servo torques and the STS3215's current, using 80 % of the pack. The model's 1.0 A of servo current while walking sits inside the range the reference robot's bench supply showed.

Safety

- A 10 A DC-rated switch between the pack and everything else; the peak is 2.52 A.
- A buzzer on the balance lead warns before the LiPo is over-discharged.
- Charge only with the balance charger, on a non-flammable surface.
- The UBEC gives the Pi 5 A; it needs up to 3 A, the audio 0.4 A.

Not included in this estimate

- A 3D printer (or a print service), and the time to print about 0.8 kg of parts
- Shipping and delivery charges from the stores
- Claude API usage (pay per use) and Wi-Fi
- Tools: PH0 / PH1 screwdrivers (one stubby), 1.5–2.5 mm hex L-keys, calipers, a soldering iron for the inserts
- A spare STS3215 (₹2,419) and a second battery, recommended
- An optional 12.6 V bench supply for long sessions at the desk

7. Complete list of items

#	ITEM	STORE	QTY	UNIT	TOTAL
01	**Joint servo (all 17 joints: 12 legs, 4 arms, neck)** Feetech STS3215 12 V 30 kg·cm serial bus servo (sold as ST3215-C018)	Evelta	17	₹2,419	**₹41,123**
03	**Bus servo driver** Waveshare Serial Bus Servo Driver board	Robu	1	₹432	**₹432**
04	**Brain** Raspberry Pi 4 Model B 2 GB (with case)	ThinkRobotics	1	₹7,000	**₹7,000**
05	**microSD card** SanDisk Ultra 32 GB A1	Quartz Components	1	₹1,499	**₹1,499**
06	**IMU** MPU6050 gyro + accelerometer	Quartz Components	1	₹127	**₹127**
07	**Microphone** INMP441 I2S MEMS microphone	Quartz Components	1	₹145	**₹145**
08	**Amplifier** MAX98357A I2S 3 W class-D amplifier	Quartz Components	1	₹124	**₹124**
09	**Speaker** Waveshare 2030 cavity speaker 8 Ω 2 W	ThinkRobotics	1	₹300	**₹300**
12	**Battery** Bonka 3S 11.1 V 2200 mAh 35C LiPo	Robocraze	1	₹2,039	**₹2,039**
13	**Charger** B3 balance charger for 2S/3S LiPo	Quartz Components	1	₹294	**₹294**
14	**5 V supply for the Pi** UBEC 5 V 5 A (2-8S input)	Robu	1	₹409	**₹409**
16	**Battery connector** XT60 male/female pair	Robocraze	2	₹29	**₹58**
17	**Power switch** 10 A DC-rated toggle switch (CALONIX)	Quartz Components	1	₹63	**₹63**
18	**Power wire** 16 AWG silicone wire (per metre)	Robu	2	₹49	**₹98**
19	**Signal wire** 20 AWG silicone wire (per metre)	Robu	3	₹21	**₹63**
20	**Battery alarm** 1-8S LiPo voltage checker / low-voltage buzzer	Quartz Components	1	₹98	**₹98**
21	**Filament for the body, head and arms (the reference robot's pastel green)** eSun PLA-Matte 1.75 mm 1 kg, Mint Green	Robocraze	1	₹1,299	**₹1,299**
21b	**Filament for the pelvis and leg brackets** PETG 1.75 mm 1 kg, green (Quartz Components PETG Pro)	Quartz Components	1	₹759	**₹759**
22	**M2 x 6 CSK screws (servo cases)** M2 x 6 SS CSK Philips	local fastener supplier	70	₹2	**₹105**
22b	**M2 x 8 CSK screws (Raspberry Pi, chest cap)** M2 x 8 SS CSK Philips	local fastener supplier	40	₹2	**₹72**
22c	**M2 x 10 CSK screws (hip-yaw keepers, head)** M2 x 10 SS CSK Philips	local fastener supplier	70	₹2	**₹133**
22d	**M2 x 12 CSK screws** M2 x 12 SS CSK Philips	local fastener supplier	10	₹2	**₹25**
23	**M3 x 8 socket screws (torso to pelvis)** M3 x 8 SS socket head	local fastener supplier	10	₹5	**₹50**
23b	**M3 x 8 pan head screws (passive horns, neck hub)** M3 x 8 SS Philips pan head	local fastener supplier	45	₹3	**₹135**
24	**M3 heat-set inserts** M3 x 4 brass heat-set inserts	local fastener supplier	25	₹4	**₹100**
25	**Foot grip** Natural rubber sheet 1 mm x 100 mm x 1 m	IndustryBuying	1	₹978	**₹978**
25b	**Hip-yaw thrust rings and keepers** PTFE thread-seal tape, 1 roll	any hardware shop	1	₹30	**₹30**
26	**Push-to-talk button** 12 mm momentary push button	any electronics shop	1	₹20	**₹20**
27	**Jumper wires and servo cables** Dupont jumpers + 4 STS/SCS servo cables, 5264 3-pin, 30 cm (Feetech /...	any electronics shop	1	₹300	**₹300**

Total for one JX0 **₹57,878**

Rupees Fifty-Seven Thousand Eight Hundred Seventy-Eight only